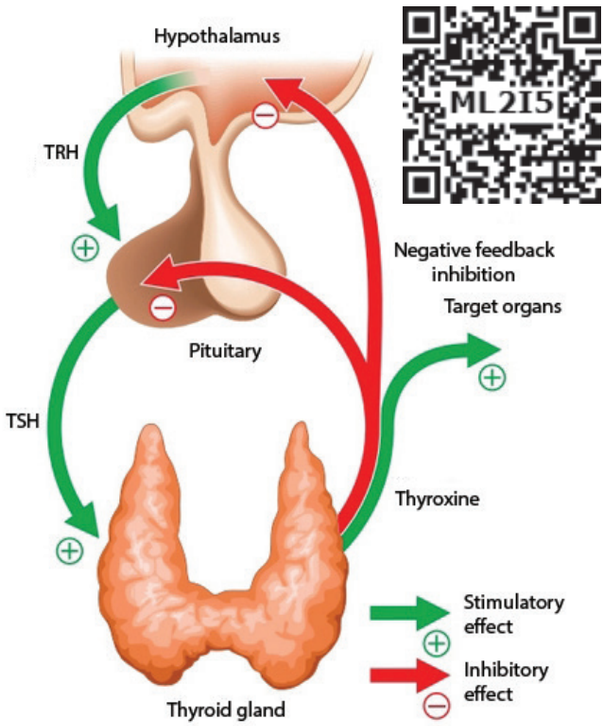


Figure 11.4 Negative feedback mechanism

Functions of thyrocalcitonin (TCT): TCT is a polypeptide hormone, which regulates the blood calcium and phosphate levels. It reduces the blood calcium level and opposes the effects of parathyroid hormone.



Iodine is required for formation of thyroxine: To produce normal quantities of thyroxine, about 1mg/week of iodine is required. To prevent iodine deficiency common table salt is iodised with 1 part sodium iodide to every 1,00,000 parts of sodium chloride

11.2.5 Parathyroid gland

In human, four tiny parathyroid glands are found in the posterior wall of the thyroid glands. This gland is composed of two types of cells, the chief cells and oxyphil cells. The chief cells secrete parathyroid hormone (PTH) and the functions of oxyphil cells are not known.

Parathyroid hormone or Parathormone (PTH)

PTH is a **hypercalcemic hormone**. It is a peptide hormone involved in controlling the calcium and phosphate homeostasis. The secretion of PTH is controlled by calcium level in the blood. It increases the blood calcium level by stimulating osteoclasts to dissolve the bone matrix. As a result calcium and phosphate are released into the blood. PTH enhances the reabsorption of calcium and excretion of phosphates by the renal tubules and promotes activation of vitamin D to increase calcium absorption by intestinal mucosal cells.

11.2.6 Thymus gland

Thymus gland is partially an endocrine and partially a lymphoid organ. It is a bilobed structure located just above the heart and aorta, behind the sternum. It is covered by fibrous capsule and anatomically it is divisible into an outer cortex and an inner medulla. It secretes four hormones such **as thymulin, thymosin, thymopoietin and thymic humoral factor (THF).** The primary function of thymus is the production of immuno competent 'T' lymphocytes which provides cell mediated immunity.

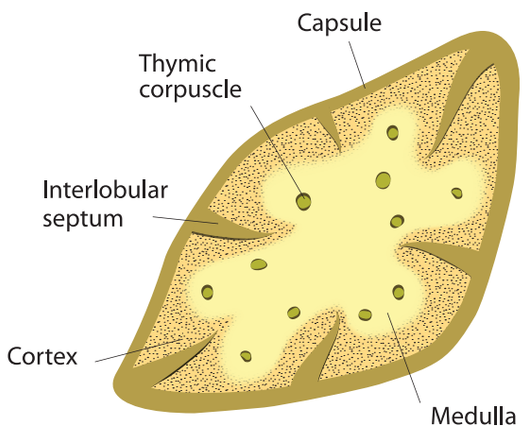


Figure 11.5 Structure of thymus gland

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